

Solving the Sensor-Based Activity Recognition Problem (SOAR): Self-Supervised, Multi-Modal Recognition of Activities from Wearable Sensors

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ABSTRACT

Feature extraction remains the core challenge in Human Activity Recognition (HAR) - the automated inference of activities being performed from sensor data. Over the past few years, the community has witnessed a shift from manual feature engineering using statistical metrics and distribution-based representations, to feature learning via neural networks. Particularly, self-supervised learning methods that leverage large-scale unlabeled data to train powerful feature extractors have gained significant traction, and various works have demonstrated their ability to train powerful feature extractors from large-scale unlabeled data. Recently, the advent of Large Language Models (LLMs) and multi-modal foundation models has unveiled a promising direction by leveraging well-understood data modalities. This tutorial focuses on existing representation learning works, from single-sensor approaches to cross-device and cross-modality pipelines. Furthermore, we will provide an overview of recent developments in multi-modal foundation models, which originated from language and vision learning, but have recently started incorporating inertial measurement units (IMU) and time-series data. This tutorial will offer an important forum for researchers in the mobile sensing community to discuss future research directions in representation learning for HAR, and in particular, to identify potential avenues to incorporate the latest advancements in multi-modal foundation models, aiming to finally solve the long-standing activity recognition problem.

CCS CONCEPTS

• **Human-centered computing** → **Ubiquitous and mobile computing**; **Empirical studies in ubiquitous and mobile computing**; • **Computing methodologies** → *Machine learning*.

KEYWORDS

human activity recognition, representation learning, self-supervision, multi-modal learning

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1 INTENDED AUDIENCE

This is an introductory to intermediate tutorial. Participants will include UbiComp and HCI researchers, industry practitioners, and students. We will expect participants to have familiarity with the basics of machine learning and Python coding. Ideal participants would have experience with coding in notebooks such as Jupyter and Google Colab.

2 PREVIOUS OFFERINGS OF THE TUTORIAL

The organizers ran related tutorials at previous conferences, which the proposed tutorial builds upon. At UbiComp 2015 Plötz (along with Anind Dey [then CMU, now UW], Mayank Goel [now CMU], Nils Hammerla [then Newcastle University]) organized a full day tutorial on “Machine Learning for Ubicomp”, which was very well attended (sold out on first day of registration). Plötz (together with Nic Lane [then Oxford and Nokia Bell Labs, now Cambridge and Samsung] and Sourav Bhattacharya [then Nokia Bell Labs, now Samsung] and Yu Guan [Newcastle University]) then organized tutorials on “Deep Learning for Ubicomp”, which were held at MobiCase 2016 and UbiComp 2017. Both tutorials were also well attended (sold out).

The previous version of this tutorial, also called ‘Solving the Sensor-Based Activity Recognition Problem (SOAR)’¹, was organized in UbiComp 2023, with great attendance [12].

3 MAIN CONTACT PERSON

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¹link: <https://sites.google.com/view/soar-tutorial-ubicomp2023/home>



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4 ORGANIZERS

Harish Haresamudram: Harish Haresamudram is a fifth year PhD student with the School of Electrical and Computer Engineering at Georgia Institute of Technology. His research involves understanding human movements and behaviors via the opportunistic use of body worn sensing platforms from wearable devices, as they enable truly in-the-wild data collection. He is particularly interested in learning representations via techniques that require minimal supervision, by leveraging large quantities of unlabeled movement data.

Chi Ian Tang: Chi Ian Tang is a research scientist in pervasive computing at Nokia Bell Labs Cambridge. He received his PhD in Computer Science from the University of Cambridge. His research focuses on developing scalable and data-efficient solutions for mobile sensing, with a special interest in leveraging massive amounts of unlabeled sensing data collected in the wild, to develop more generalizable models for mobile sensing in order to provide deeper insights about user behaviors. His work has appeared in ICML, UbiComp (ACM IMWUT), and ICASSP, among others, and he has been a reviewer for IMWUT, ISWC and ACM HEALTH.

Sungho Suh: Sungho Suh is a Senior Researcher at the German Research Center for Artificial Intelligence (DFKI) in Germany. He received the Ph.D. degree in Computer Science at the Technische Universität Kaiserslautern, Germany in 2021, and the B.S. and M.S. degrees from the School of Electrical Engineering and Computer Science, Seoul National University, Seoul, South Korea, in 2009 and 2011, respectively. His research interests are machine learning algorithms, such as sensor data processing, computer vision, multi-modal processing, and generative models, with a focus on industrial applications.

Paul Lukowicz: Paul Lukowicz is both Scientific Director at the German Research Center for Artificial Intelligence (DFKI GmbH) and Professor of Computer Science at RPTU Kaiserslautern-Landau in Germany since 2012 where he heads the Embedded Intelligence group. His research focuses on context-aware ubiquitous and wearable systems including sensing, pattern recognition, system architectures, models of large-scale self-organized systems, and applications in areas ranging from healthcare through industry 4.0 to smart cities.

Thomas Plötz: Thomas Plötz is a computer scientist with expertise and almost 20 years of experience in Pattern Recognition and Machine Learning research (Ph.D. from Bielefeld University, Germany). His research agenda focuses on applied machine learning, that is developing systems and innovative sensor data analysis methods for real world applications. Primary application domain for his work is computational behavior analysis where he develops methods for automated and objective behavior assessments in naturalistic environments. Main driving functions for his work are "in the wild" deployments and as such the development of systems and methods that have a real impact on peoples' lives. Thomas is currently Associate Professor at the School of Interactive Computing at Georgia Institute of Technology.

5 MOTIVATION / RELEVANCE TO THE COMMUNITY

Representations used for sensor-based HAR have evolved over time. Traditionally, statistical metrics and distribution-based features were used as the descriptors of movements present in windows of sensor data, in conjunction with simpler classifiers [1, 18]. However, advancements in Deep Learning (DL) and its adoption by the Ubi-comp community, allowed practitioners and researchers to leverage the publicly available annotated datasets to directly *learn* representations (i.e., neural network parameters) that are useful for solving the specific HAR task [8, 17]. However, the lack of large-scale annotated datasets (of thousands of hours/users) limits the scale and complexity of the networks that can be applied [11]. This leads to most supervised deep networks being relatively smaller in size (compared to other domains like vision and natural language) [14]. In this tutorial, we will first present the Activity Recognition Chain (ARC) [1], which is the framework for recognizing activities, and discuss the drawbacks and advantages of the traditional, statistical features. Furthermore, we will present DeepConvLSTM [17], which is the established supervised deep learning method for HAR, to contrast against heuristic-based features as well as learning without any label information.

Simply *collecting* large-scale sensor data without annotations is more straightforward, and data has previously been collected from a couple hundred to thousands of participants, e.g., in the Capture-24 [2, 3] and UK Biobank studies [6]. Such datasets have empowered research into 'Self-Supervised' learning, which performs feature learning without the need for annotations [7, 11]. It comprises two stages: (i) pre-training, where pretext tasks are designed to capture specific aspects of input sensor data; and (ii) classification, where the features are used to recognize activities. A number of such methods have been designed specifically for time-series wearable sensor data, e.g., multi-task probabilistic transformation discrimination [20] and reconstruction of data [9]. Another family of methods involves contrastive learning, which has been successfully applied through random augmentations [19, 21] and through future timestep prediction [10]. Currently, self-supervised methods are the defacto for recognizing activities. Therefore, we will also discuss and present Autoencoders (which are simple to setup and are therefore an established baseline), as well as SimCLR (involving contrastive learning across differently augmented versions of the same input).

Further, the wearables community has been developing multi-sensor [5, 13, 16] and multi-modal methods for the recognition of activities [4, 15]. Going beyond utilizing wearable sensors in multiple on-body locations, these approaches aim to use entirely different sensing modalities, such as cameras, audio, and more recently, natural language. These recent efforts towards learning joint embedding spaces with natural language have been in pursuit of developing *foundational* models of human movement. This mirrors the trends in the machine learning community in general, where there is substantial interest in developing foundational Large Language Models (LLMs). In light of this, our tutorial will also cover such multi-modal methods, with a particular emphasis on foundational methods. We will present the current status in the field, followed by the areas of opportunity.

Table 1: Tentative schedule for the tutorial.

Time	Block	Presenter
8:30 AM - 9:00 AM	Registration and Setup	All
9:00 AM - 9:15 AM	Start: Welcome, Introduction, Overview of HAR and its development.	Thomas/Paul
9:15 AM - 10:00 AM	Introduction to self-supervised learning methods for wearable sensor data (with hands-on exercises).	Harish/Sungho/Ian
10:00 AM - 10:45 AM	Contrastive learning leveraging multiple sensors and sensor modalities (with hands-on exercises).	Harish/Sungho/Ian
10:45 AM - 11:15 AM	Overflow time and brief break	–
11:15 AM - 12:00 PM	Recent developments in multimodal foundation models; aligning HAR models to language and vision models.	Harish/Sungho/Ian
12:00 PM - 12:30 PM	Forum discussion about the state-of-the-field; audience thoughts on promising future methods; Conclusion.	Thomas/Paul

Finally, we will have an open discussion regarding the state of the field, and we will debate, discuss, and present the most pressing problems in the field, as well as the potential lines of new research. Overall, this tutorial is a revised version of our previous offering in UbiComp 2023, with the increased focus on multi-modal and foundational models, following the trends in the community. We aim to bring together researchers and experts alike, in order to gauge the progress of the field towards solving the activity recognition problem.

6 BRIEF OUTLINE OF THE TOPICS TO BE COVERED

- Introduction to HAR and challenges involved in sensor-based HAR; sensor data analysis, including the Activity Recognition Chain (ARC).
- Unsupervised and self-supervised learning methods for wearable-based human activity recognition.
- Contrastive learning methods, including cross-device approaches.
- Recent developments in multi-modal foundation models and their application to time-series and sensor data.

7 FORMAT AND DETAILED SCHEDULE

The tutorial is planned as a half-day event that focuses on three main components:

- (1) Presentations by the organizers.
- (2) Hand-on exercises.
- (3) Open discussion regarding the methods, the field, and its future.

The detailed (tentative) breakdown of the tutorial can be found in Tab. 1.

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